What You See Is What You Get: Exploring Gen Z Consumers' Attitudes Towards Digital Labelling and QR Codes in Apparel

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***Abstract*—Apparel labels have traditionally been small sewn-in tags that carry limited information on fibre content, care and origin. Digital labelling, most often implemented through Quick Response (QR) codes printed or woven into garments, links a single garment to rich and updatable product information such as materials, supply chain, sustainability credentials, care guidance and recycling instructions. Generation Z is the first cohort to grow up fully immersed in smartphones and is also widely described as sustainability-conscious, yet its willingness to rely on digital labels remains unclear. This paper examines Gen Z attitudes towards digital labelling and QR codes in apparel through a structured synthesis of recent published evidence, including a national consumer survey, an international industry survey, and academic studies on digital clothing labels. Using the Technology Acceptance Model and the Theory of Planned Behaviour as a lens, we propose a conceptual model and assess six propositions against the available evidence. The evidence shows strong interest in digital information about care, authenticity and recycling, but also a clear preference for keeping printed labels, and a gap between comfort with QR codes and frequent use. Evidence on privacy concern is largely missing. The paper closes with design recommendations for brands and engineers and a research agenda for primary data collection.**

***Index Terms*—digital labelling, QR codes, apparel, Generation Z, technology acceptance, consumer attitude, sustainability, transparency**

I. Introduction

Clothing labels exist to inform. Regulations in many regions require garments to declare their fibre composition and care instructions, and brands add details such as size, country of origin and brand identity [12]. However, the physical space on a sewn-in tag is extremely limited, and the text is often small, multilingual and difficult to read. At the same time, consumers increasingly ask for information that a tag cannot hold, such as where a garment was made, what its environmental footprint is, whether it can be repaired or recycled, and how it should be disposed of at end of life [10].

Digital labelling addresses this gap. A QR code, standardised in ISO/IEC 18004, can be printed, woven or heat-transferred onto a garment and scanned with any modern smartphone camera. The code resolves to a web page or application that can show unlimited, multilingual and updatable content. In Europe, the Ecodesign for Sustainable Products Regulation introduces the concept of a Digital Product Passport, which makes machine-readable product data a regulatory direction for textiles [11]. Digital labels are therefore moving from a marketing experiment towards a likely industry norm.

The success of this shift depends on consumers. A digital label is useful only if people scan it, understand it and trust what it shows. Generation Z, commonly defined as those born from 1997 onwards [9], is a particularly important group because these consumers are comfortable with mobile technology and frequently state that sustainability and transparency matter to them. Yet comfort with technology does not automatically translate into acceptance of QR codes on clothing.

The phrase "what you see is what you get" captures the expectation that a label should honestly show what is in a product. A recent qualitative study of 63 university students has explored Gen Z perceptions of digital labels directly [1]. Building on that work and on other recent surveys, this paper makes three contributions:

- a conceptual model of Gen Z attitude and intention towards digital labels that integrates technology acceptance, trust, sustainability awareness and privacy concern;
- a structured comparison of published evidence against six propositions derived from the model; and
- design recommendations for brands and for engineers building QR-based label systems, with a research agenda.

The remainder of the paper is organised as follows. Section II gives background. Section III develops the propositions. Section IV describes the evidence-synthesis method. Section V presents the evidence and results. Section VI discusses implications and limitations, and Section VII concludes.

II. Background

A. Digital Labelling and QR Codes in Apparel

A digital label links a physical garment to online information through a machine-readable identifier. The identifier may be a QR code, an NFC tag or an RFID tag, but QR codes dominate because they are inexpensive, need no special hardware in the garment and are readable by nearly every smartphone. Fig. 1 shows the typical interaction. The garment

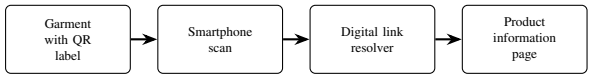


Fig. 1. Typical interaction flow of a QR-based digital label.

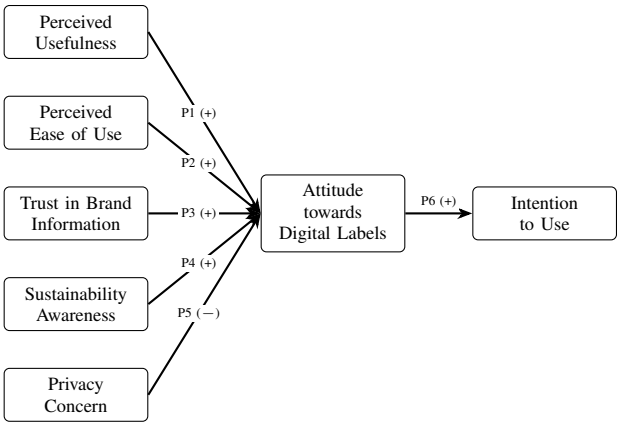


Fig. 2. Proposed conceptual model.

carries a code, the consumer scans it, a resolver maps the identifier to the correct product record, and the brand presents information such as material composition, care instructions, supply chain data and end-of-life options.

Compared with a sewn-in tag, a digital label offers greater information capacity, the ability to update content after production, multilingual delivery from one garment, and a channel for engagement such as repair and resale. It also introduces drawbacks: it needs a device and a connection, adds a step for the consumer, raises data privacy questions, and can fail if the brand's link is removed or broken.

B. Theoretical Foundations

We draw on three established theories. The Technology Acceptance Model (TAM) proposes that perceived usefulness and perceived ease of use determine attitude and intention to use a technology [6]. The Theory of Planned Behaviour (TPB) holds that attitudes, subjective norms and perceived control predict behavioural intention [7]. The Unified Theory of Acceptance and Use of Technology (UTAUT) integrates several acceptance models [8]. Our model keeps the core TAM structure and extends it with trust, sustainability awareness and privacy concern, which are especially relevant to product information and personal data.

III. Conceptual Framework and Propositions

Fig. 2 presents the proposed model. Five antecedents influence Attitude towards Digital Labels (ATT), and ATT influences Intention to Use Digital Labels (INT). Because this paper relies on published evidence rather than new respondent data, the relationships are stated as propositions (P1 to P6) that are assessed in Section V.

P1 (Perceived usefulness). The more consumers believe digital labels give them better information for buying and caring for clothes, the more favourable their attitude.

P2 (Perceived ease of use). The more effortless scanning and reading the linked content feels, the more favourable the attitude.

P3 (Trust). The more consumers trust that the information behind the code is accurate and honest, the more favourable the attitude. Because a QR code can lead anywhere, trust is central to the "what you see is what you get" promise.

P4 (Sustainability awareness). Consumers who care about environmental and social impact value the extra information that digital labels provide and hold more favourable attitudes.

P5 (Privacy concern). Worry that scanning a code exposes personal data weakens the attitude towards digital labels.

P6 (Attitude and intention). A more positive attitude leads to stronger intention to use digital labels.

IV. Methodology

A. Approach

The study is a structured narrative synthesis of published evidence. Instead of collecting new respondent data, we gathered recent consumer surveys and academic studies that report attitudes towards QR codes or digital labels on clothing, extracted their sample sizes and headline findings, and compared them with the six propositions.

B. Source Selection

Sources were identified through web searches using the terms digital labelling, QR codes, apparel, clothing labels and Gen Z. A source was included if it (i) reported empirical findings on consumer views of clothing labels, QR codes or digital labels, (ii) stated its sample and method, and (iii) was published between 2023 and 2026. Five sources met these criteria (Table I). Findings are reported as published, and only claims stated by the source are used.

C. Assessment of Propositions

Each proposition was rated as *supported*, *partly supported*, *mixed* or *not tested*, depending on whether at least one source provided direct or indirect evidence for or against it. Because the sources differ in design and population, no statistical pooling was attempted.

V. Evidence and Results

A. Evidence Base

Table I summarises the sources. Only one source focuses specifically on Gen Z [1]; the others cover general consumers, so Gen Z findings must be interpreted with care.

B. Key Patterns

Interest in digital information is real, but conditional. Industry surveys show that a majority in several markets find QR-based care and recycling information appealing [3], [4]. Appeal varies by region, from 72% in China to 48% in Europe [4].

Printed labels remain the default. In the U.S. survey, three quarters of consumers preferred printed labels to QR codes [2]. Fig. 3 shows the gap between comfort with QR codes and

TABLE I
SUMMARY OF PUBLISHED EVIDENCE USED IN THIS STUDY

Source	Sample / method	Population	Key finding (paraphrased)
Ghalachyan and Loranger [1]	63 students, two universities; online questionnaire, qualitative	Gen Z (U.S.)	Explores experiences with traditional labels and QR codes, information consumers want beyond traditional labels, attitudes to digital labelling, and views on how brands could use QR technology, framed as consumer empowerment.
Cotton Inc. [2]	978 consumers; survey, July 2023	U.S. adults, all generations	Most consumers value printed labels: 75% prefer printed labels to QR codes and 22% prefer QR codes. 66% are comfortable using QR codes but only 31% use them frequently. Size (83%) and care (80%) are the most helpful items. Among Gen Z, 33% would buy a different item without labels and 56% expect a laundering mishap.
Avery Dennison and GWI [3]	Over 6,300 clothing shoppers, seven countries	Fashion shoppers	About six in ten consider scanning a QR code for garment care a valuable tool, and comfort with digital triggers on garments increased compared with the previous year's report.
Avery Dennison and GWI [4]	Over 5,000 fashion consumers, five markets; concept testing	Fashion shoppers (U.S., UK, France, Germany, China)	Scanning QR codes for care and recycling information was appealing to 72% in China, 55% in the U.S. and 48% in Europe. 71% found a digital authenticity check appealing.
Kim, Cho and Fiore [5]	302 consumers; survey, ELM and diffusion theory	U.S. adults	Compatible advantage, observability and green transparency raised perceptions of brand technological expertise; accessibility had no significant effect; brand technological expertise raised perceived brand green value.

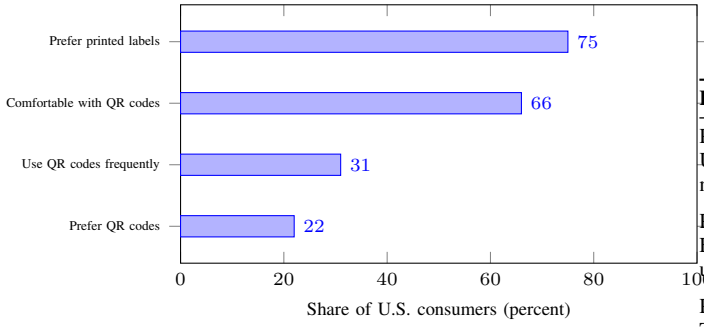


Fig. 3. U.S. consumer views on printed labels and QR codes (data from Cotton Incorporated, N = 978) [2].

frequent use or preference for them. This points to an attitude-behaviour gap: people accept the technology in principle but do not yet see it as a replacement for the tag.

Practical value drives appeal. Care guidance is the most valued label content [2], and care information through QR codes is seen as valuable for making clothes last longer [3]. Gen Z respondents were the group most likely to expect a laundering mishap without labels (56%), which suggests that care information is especially relevant to them [2].

Digital labels shape brand perception. In the academic study, perceived advantages and green transparency improved how consumers judged a brand's technological expertise, which in turn improved perceived green value [5].

C. Assessment of the Propositions

Table II compares the propositions with the evidence.

TABLE II
ASSESSMENT OF PROPOSITIONS AGAINST PUBLISHED EVIDENCE

Prop.	Evidence	Rating
P1 Usefulness	Care and recycling information seen as valuable [3], [4]; relative advantage raised brand perceptions [5]	Supported
P2 Ease of use	High comfort but low frequent use [2]; accessibility had no significant effect [5]	Mixed
P3 Trust	Authenticity check appealing to 71% [4]; green transparency raised brand perception [5]	Partly supported
P4 Sustainability	Gen Z described as invested in sustainability and transparency [1]; brand green value effect [5]	Partly supported
P5 Privacy	Not measured in the reviewed sources	Not tested
P6 Attitude to intention	Rising comfort [3], but strong preference for printed labels [2]	Partly supported

VI. DISCUSSION

A. Interpretation

Three conclusions emerge. First, usefulness is the clearest driver: consumers respond to concrete benefits such as care guidance, authenticity checks and recycling advice rather than to the technology itself. Second, ease of use is a more complicated factor. Comfort with QR codes is high, but frequent use is much lower, so the extra step of scanning remains a barrier at the moment of purchase or laundering. Third, none of the reviewed sources tested privacy concern, which is a notable gap given that a scan can reveal device and location data.

B. Practical Implications

- **Keep a physical fallback.** Retain legally required fibre and care information on a sewn-in tag and use the QR code for additional content, since most consumers still prefer printed labels [2].

- **Lead with care and recycling.** Show the most valued facts first on the landing page, in a fast mobile-friendly layout.
- **Build trust visibly.** Offer verifiable data, an authenticity check and a recognisable brand domain.
- **Respect privacy by design.** Avoid mandatory sign-up, minimise tracking and explain what data is collected on scan.
- **Engineer for durability.** Codes should survive washing, remain scannable on dark or textured fabric, and resolve through a persistent link so old garments keep working.

C. Implications for Computer Engineers

Digital labelling is also a software and systems problem. Reliable code generation and error correction, persistent resolver services, scalable product databases, secure APIs, accessibility for users with visual impairments, and protection against malicious QR codes ("quishing") are all engineering challenges. A prototype label resolver that measures scan success on printed and woven fabric would be a natural next step from a computing perspective.

D. Limitations

This study depends on secondary evidence. Only one reviewed source is specific to Gen Z and it is small and qualitative [1]. Two sources are industry surveys that may reflect commercial interests, and the sources differ in countries, years and question wording, so results are not directly comparable. No statistical testing of the model was possible.

E. Future Research

Future work should test the model with a primary survey of Gen Z consumers using multi-item scales (see the Appendix), include privacy concern explicitly, compare Gen Z with other generations, include Indian consumers, who are absent from the reviewed sources, and observe actual scanning behaviour in field experiments.

VII. Conclusion

Digital labelling through QR codes offers apparel brands a way to give consumers richer, more current and more transparent information than a physical tag can hold. The published evidence shows that consumers value useful digital information, especially on care, authenticity and recycling, but most still prefer printed labels and do not yet use QR codes frequently. For Gen Z, who are digitally fluent and care about transparency, the key to acceptance is therefore not the technology itself but clear usefulness, trustworthy content, effortless access and respect for privacy. Brands that combine a physical label with a fast, honest and privacy-respecting digital layer are best placed to ensure that what the consumer sees on the label is what they get in the product.

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Appendix A
Proposed Questionnaire for Future Primary Research

All items use a five-point Likert scale (1 = strongly disagree, 5 = strongly agree) and should be pilot tested before use.

TABLE III
Proposed Measurement Items

Construct	Item
PU	A QR code on clothing gives me more useful information than a normal tag. QR labels help me look after my clothes properly.
PEU	Scanning a QR code on clothing is easy for me. The information I get after scanning is easy to understand.
TR	I believe the information behind a clothing QR code is accurate. I feel confident that a QR label leads to a genuine brand page.
SA	I want to know how and where my clothes were made. I consider recycling or disposal when buying clothes.
PC	I worry that scanning a QR code may reveal my personal data. I am concerned that brands may track me through QR labels.
ATT	I have a positive view of digital labels on clothing. Digital labelling is a good idea for the apparel industry.
INT	I intend to scan QR labels on clothing in the future. I would prefer to buy from brands that offer digital labels.